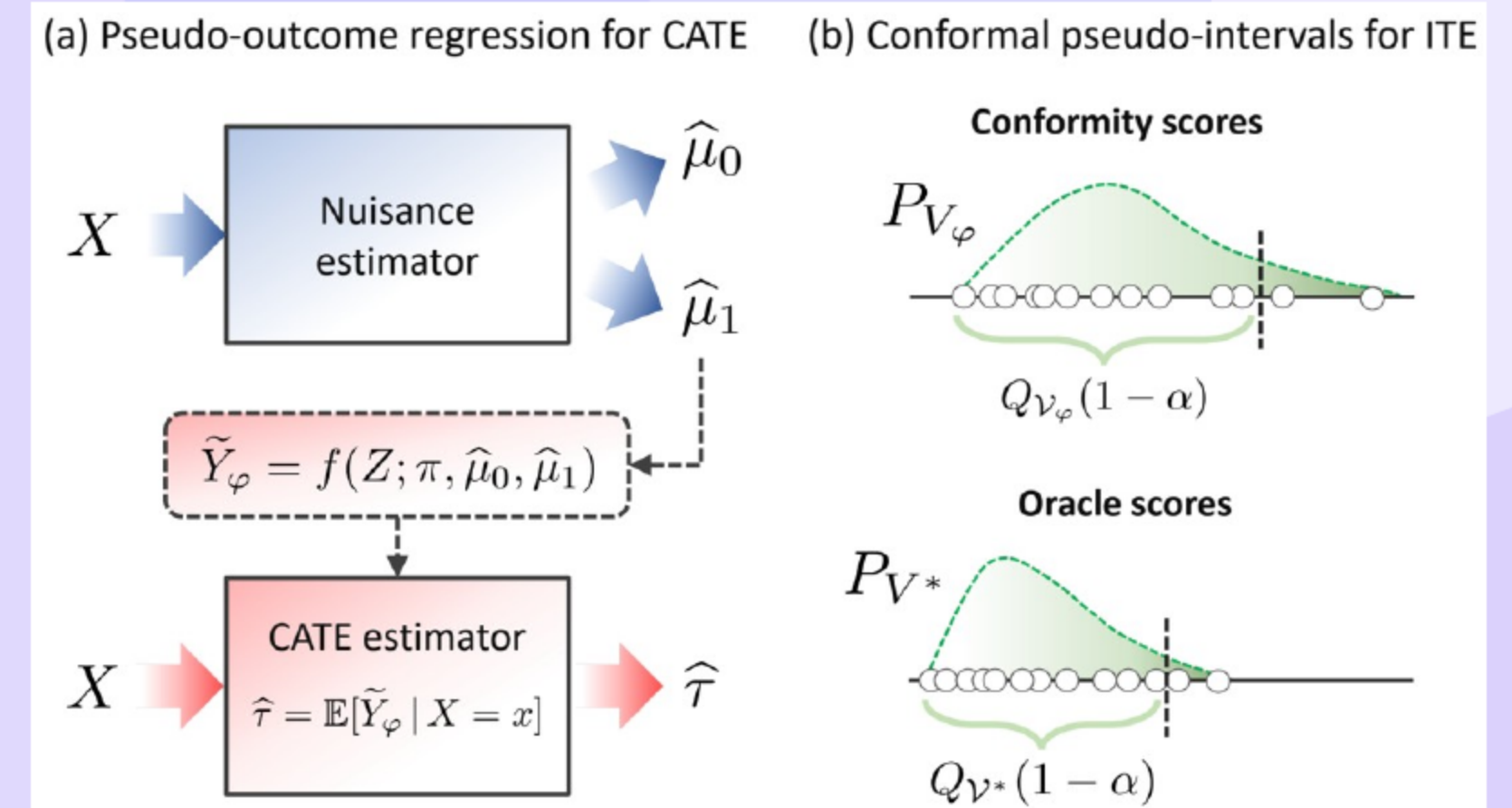


# Conformal Meta-learners for Predictive Inference of Individual Treatment Effects

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## 01 Visual Overview: Conformal Meta-learner Pipeline



- Post-hoc \Conformal over meta-learners
- Split: nuisances  $\rightarrow$  pseudo-outcomes
- Calibrate scores  $\rightarrow$  ITE \intervals

## 02 Validity Insight (Combined)

- **Validity:** pseudo-score dominance  $\Rightarrow$  coverage
- FOSD/SOSD/MCX link  $V_{\text{link}}$  vs oracle  $V$
- **IPW/DR** dominate; X-learner lacks guarantees

## 03 One-slide Comparison vs Exact Nested WCP (Baseline)

- **Baseline:** Exact Nested WCP
- At 90%, DR/IPW cover with shorter intervals
- **Takeaway:** Matches coverage, shorter intervals than Exact Nested WCP

# Thank You

Questions & Discussion